SANJANA RANI

Aerospace Engineer | Gas Turbine Aerodynamics & CFD | Python/MATLAB

+91-9996673079 | sanjanakamboj98@gmail.com | linkedin.com/in/sanjana-rani | github.com/Sanjanakamboj

Date: September 4, 2026

Mr. Eric Blidmark, GTC Engineering Manager
GKN Aerospace – Global Technology Center, Future Concepts
Trollhättan, Sweden

Subject: Application for Aerodynamics Engineer – Future Concepts, GTC

Dear Mr. Blidmark,

I am writing to apply for the Aerodynamics Engineer position in the Future Concepts team at GKN Aerospace's Global Technology Center. I hold a joint international Master's degree in Transfer of Fluid Materials in Aeronautical and Space Applications (TFM-ASA) across Université de Bordeaux, UCLouvain, and BTU Cottbus, with hands-on experience across multiple fidelity levels of turbomachinery aerodynamic design.

I designed and validated an axial turbine stage from 1D mean-line thermodynamics through 2D compressible RANS simulation (SU2, GMSH, ParaView) – the same mean-line-to-CFD progression this role's methodology spans – and built a self-directed OpenFOAM practice covering turbulence-model comparisons and mesh generation. My ANSYS Fluent experience comes from a Bachelor's final project and a freelance internship at Simulation Lab studying airfoil performance across angle-of-attack sweeps. Separately, I built a from-first-principles CFD study of cylinder flow, cross-validating a from-scratch Navier-Stokes solver and an OpenFOAM mesh-independence study against published Strouhal-number and drag-coefficient data.

Across all of this work, my consistent habit is to validate every result against an analytical solution, published data, or an independent method before trusting it – the same rigor I understand this role expects when moving between fidelity levels and design stages.

I want to be upfront about a few things: my Swedish is not yet at a working level, though I am fluent in English and motivated to build Swedish proficiency given the opportunity. My hands-on CFD experience has been in aerodynamic and fluid-dynamic analysis rather than the aeroacoustics, aeroelastic (flutter/forced response), or heat transfer analyses this role also involves, and I have not used UG-NX or worked with Model-Based Systems Engineering. I see these as genuine areas to grow into, backed by a strong foundation in thermodynamics, aerodynamics, and CFD methodology.

I have enclosed my CV and would welcome the opportunity to discuss how my background fits the team's needs. Thank you for your consideration.

Sincerely,
Sanjana Rani